A Survey on Multimodal Deep Learning for Image Synthesis

Applications, methods, datasets, evaluation metrics, and results comparison

Sanbi Luo
luosanbi@iie.ac.cn
School of Cyber Security, University of Chinese Academy of Sciences, Beijing
China
Institute of Information Engineering, Chinese Academy of Sciences, Beijing
China

ABSTRACT

In recent years, deep learning technology has made breakthroughs in computer vision. After using large-scale data training, the deep neural network represented by GAN is significantly better than previous technologies in image generation, including generating more reasonable, higher-definition, more complex, and more accurate images. With the continuous development of datasets, models, and applications, the fusion of different modal information, including fusion of natural language, semantic layouts, tags, edge maps, and other different modal information, to generate images has become a new demand and challenge. There are related reviews on image generation and multimodal deep learning. However, there has not been a review dedicated to multimodal deep-learning image generation to discuss the current status, existing problems, and challenges of this task. Therefore, this review proposes a survey on multimodal deep learning image generation. It aims to provide readers with an application scenario for multimodal deep learning image generation. Also, it provides readers with new multimodal deep learning image generation technologies, the relevant datasets, evaluation metrics used, and some results comparison. Finally, this article describes some of the challenges and future topics of multimodal deep learning image generation.

CCS CONCEPTS

• **Networks** → **Network architectures**; **Network performance evaluation**; • **Computing methodologies** → **Artificial intelligence**; **Computer vision**.

KEYWORDS

image synthesis, multimodal, deep learning, machine learning, computer vision

ACM Reference Format:
Sanbi Luo. 2021. A Survey on Multimodal Deep Learning for Image Synthesis: Applications, methods, datasets, evaluation metrics, and results comparison. In *2021 the 5th International Conference on Innovation in Artificial Intelligence (ICIAI 2021), March 5–8, 2021, Xia men, China.* ACM, New York, NY, USA, 13 pages. https://doi.org/10.1145/3461353.3461388

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICIAI 2021, March 5–8, 2021, Xia men, China
© 2021 Association for Computing Machinery.
ACM ISBN 978-1-4503-8863-4/21/03…$15.00
https://doi.org/10.1145/3461353.3461388

1 INTRODUCTION

The way humans perceive the world is multimodal — we see colors, hear sounds, feel texture, taste flavors, smell odors, etc. For a long time, computers have always tried to process multimodal information like humans as an extension of human brains. Especially in recent years, with the rapid development of deep learning in computer vision, NLP, etc., many technologies based on multimodal deep learning have emerged, such as multimodal data fusion [22] and multimodal representation learning [87]. Generally speaking, multimodal information is a more fine-grained concept than multimedia. Different multimodal information may exist under the same medium. For example, semantic layouts, line arts, and original images are different modalities presented by the same image. Of course, different multimedia information also belongs to the category of multimodal information. For example, images, videos, audios, and text are different modal information.

There are many review papers surveying and summarizing a large number of multimodal deep learning and image process methods. In [5], authors extensively compare more than 30 state-of-the-art super-resolution Convolutional Neural Networks (CNNs) over three classical challenging datasets to benchmark single image super-resolution. In [76], authors introduce the recent research on GANs [23] in the field of image processing, including image synthesis, image generation, image semantic editing, image-to-image translation, image super-resolution, image inpainting, and cartoon generation. In [1], authors review the most recent development in the text-to-image synthesis research domain. In [73], authors provide a comprehensive list of the existing image synthesis methods for visual machine learning. In [9], authors survey the recent advances in multimodal machine learning itself and present them in a common taxonomy. However, all the surveys mentioned above did not specifically investigate and analyze the subject of multimodal deep learning image generation.

This survey discusses the applications, methods, datasets, and evaluation metrics of recent multimodal deep learning image generation models in the five most representative fields: text to image, scene graph to image, semantic layout to image, text-based colorization, and multimodal medical image synthesis. Particularly, we define multimodal deep learning image generation problem as:

Problem Formulation: Multimodal deep learning image synthesis aims to generate a target image through a deep learning model under at least one input that is different from the target

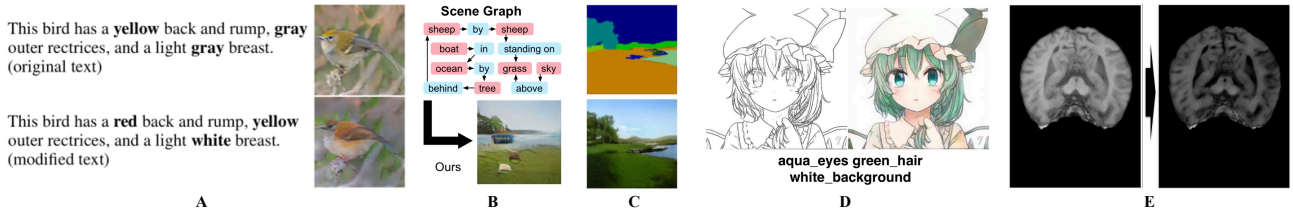

Figure 1: Some typical multimodal deep learning image synthesis applications. A denotes text to image [41], B denotes scene graph to image [34], C denotes semantic layout to image [99], D denotes text-based colorization [37], and E denotes multimodal medical image synthesis [45].

generated image modal. Accordingly, a trivial formulation can be written as:

$$I_g = G(\sum_{i=1}^{n\geq 1} I_i + \alpha I_o) \tag{1}$$

I_g denotes the target image to be generated. $G()$ denotes image generator. I_i denotes an input, which is different from I_g's modality, where i represents the i-th modal type, and the value n is not less than 1. I_o represents an original image input, which belongs to the same modality as I_g , and the coefficient α takes the value 0 or 1. According to this formulation, cross-modal generation, for example, no matter what kind of text to image generation, will definitely belong to multimodal image generation, which will help eliminate the ambiguity of multimodal attribution. In summary, this report provides the following contributions:

1. A formalization of the multimodal deep learning image synthesis problem, making the multimodal deep learning image synthesis task clearer.
2. A classification of the current image synthesis applications of multimodal deep learning.
3. A review of the architecture of typical models designed for multimodal deep learning image synthesis.
4. An overview of typical datasets for multimodal deep learning image generation.
5. An introduction of mainstream evaluation metrics of multimodal deep learning image synthesis.
6. A summary of performance of different methods with respect to the most widely used datasets and metrics.

2 APPLICATIONS

There are various multimodal deep learning image synthesis applications. This paper only reviews the five most typical of them: text to image generation, scene graph to image generation, semantic layout to image generation, text-based colorization, and multimodal medical image generation, as shown in Figure 1.

2.1 Text to Image Generation

Text-to-image synthesis refers to a computation method that transforms natural language descriptions into images with semantics similar to the descriptions. In earlier studies, image synthesis mainly relied on the correlation analysis between words and images and combined with supervision methods to find the best alignment of visual content matching the text. With deep learning development, especially the emergence of deep generative models, these models can use appropriately trained neural network models to generate realistic visual images. The text-to-image synthesis has aroused great interest in the industry, such as virtual reality, entertainment, games, computer-aided design, etc. As shown in A in Figure 1, the multimodal deep learning text to image generation model [41] generates two birds with these different color characteristics based on two sentences describing the different colors of a bird's back, rump, outer rectrices, and breast. Another famous case is that OpenAI trained a neural network this year called DALL·E [58] that creates images from text captions for a wide range of concepts expressible in natural language. It has a diverse set of capabilities, including creating anthropomorphized versions of animals and objects, combining unrelated concepts in plausible ways, rendering text, and applying transformations to existing images.

2.2 Scene Graph to Image Generation

Scene Graphs represent scenes as directed graphs, where nodes are objects and edges give relationships between objects. Scene graphs can generate complex images with many objects and relationships more efficiently than natural language sentences. Since the information conveyed by complex sentences can usually be more clearly expressed as scene graphs of objects and their relationships, scene graphs are often used as an adequate substitute for sentences to generate more appropriate image content. B of Figure 1 shows the multimodal deep learning scene graph to image generation model [34] which generates realistic images based on a scene graph derived from a sentence.

2.3 Semantic layout to Image Generation

Image semantic segmentation (semantic layout) simplifies the representation of an image into color segments that are easier to analyze. Therefore, an image can be generated from the semantic layout more accurately, at least from the spatial perspective. The generation process is also more controllable, e.g., changing a segment's color and size to affect the final generation result. C in Figure 1 shows a multimodal deep learning image synthesis model [99] generates an image from a semantic label. The model generates corresponding objects according to the different color segments in the semantic layout. For example, blue represents the sky, and yellow represents the grass. If one manipulates the segments in the semantic layout, the output image will change as well.

2.4 Text-based Colorization

Image colorization is an essential branch of image processing and computer vision to colorize images. Recently, deep learning technology has made significant progress in image colorization. Many

Table 1: A summary of different text to image generation models

Methods	Input	Output	Characteristics	Resolution
GAN-INT-CLS [60]	text	image	——	64×64
GAWWN [59]	text + location	image	location-controllable	128×128
ChatPainter [65]	text + dialogue	image	diversity	128×128
HDGAN [92]	text	image	high quality and resolution	512×512
TAC-GAN [17]	text	image	diversity	128×128
AttnGAN [85]	text	image	high quality and high inception score	256×256
StackGAN [88]	text	image	high quality	256×256
StackGAN++ [89]	text	image	high resolution photo-realistic	256×256
alignDRAW [49]	text	image	use rnn	32×32
ControlGAN [41]	text	image	high quality and controllable	256×256
Net2Net [61]	text	image	conditional invertible	256×256
OP-GAN [29]	text	image	a new evaluation metric called SOA	256×256
Multi-obj-GAN [28]	text	image	location control	256×256
DF-GAN [69]	text	image	simpler architecture + efficient	256×256
DM-GAN [97]	text	image	a dynamic memory module to refine image contents	256×256
GeNeVA [19]	text	image	recurrent generation	128×128
LeicaGAN [56]	text	image	unified framework to combine multi-phases	256×256
MirrorGAN [57]	text	image + text	semantic consistency	256×256
TediGAN [82]	text + image	image	image manipulation by text	1024×1024
VICTR [25]	text	image	visual context text representation	256×256
BilinearGAN [50]	text + image	image	manipulation by text	256×256
e-AttnGAN [2]	text	image	enhanced AttnGAN	256×256
ISLH-Txt2Im [31]	text	image	High inception score	128×128
FiLMedGAN [24]	text + image	image	manipulation by text	128×64

applications have emerged, such as gray-scale colorization, line art colorization, and real-time colorization. One of the most typical multimodal image generations is text-based colorization, which is that the image colorized based on the input text description. D in Figure 1 shows the multimodal deep learning text-based colorization model [37] colorizes a line art image into a colorized image consistent with the text tag description.

2.5 Multimodal medical Image Generation

In medical imaging, such as magnetic resonance imaging (MRI), you can use different imaging settings (such as T1, T2, DWI, Flair) to obtain several images, multimodal representations of the same image. These multimodal representations (images) have similar anatomical structures but different contrasts, providing much information for diagnosis. However, due to scan time limitations or damage caused by noise, images under specific imaging settings may not be usable. Some settings are made from the available MR images to derive some different modalities. Multimodal image generation, multi-scale feature maps from different input modalities are merged to generate target images, which is the generation of medical images based on multimodal deep learning. E in Figure 1 shows a multimodal MR image synthesis application [45]. The application can take the multimodal information extracted from MR images to generate required target images.

3 METHODS

In this section, we introduce representative multimodal deep learning image generation models, including the existing text to image generation, scene graph to image generation, semantic layout to image generation, text-based colorization, and multimodal medical image generation models.

3.1 Text to Image Generation

The text-to-image synthesis is a typical application of multimodal deep learning. If the algorithm can generate truly realistic images from only text descriptions, we can be highly confident that it can understand not only text description but also image content. Multimodal deep learning aims to teach computers to perceive and understand multimodal information and respond correctly like humans.

From Table 1, we observe that the existing text-to-image generation models focus on the realistic-looking, accuracy, operability, and high-resolution of result images. Generally, the input is a simple text sentence, and the output is a vivid image. The text has two primary purposes. The first is the only source input to provide semantic information for image generation. The second is that the text is used as a conditional input to change the image's content and color in image manipulation tasks. Most of the current images can only generate 256×256 resolution realistic-looking images.

Table 2: A summary of different scene graph (or layout) to image generation models

Methods	Input	Output	Characteristics	Resolution
sg2im [34]	scene graph	image	first paper	64×64
Layout2Im [93]	bounding boxes + object categories	image	Generate a set of realistic images	64×64
Interactive-sg2im [52]	scene graph	image	modification	——
USGC2im [71]	scene graph	image	two new metrics	64×64
PasteGAN [42]	scene graph + image crops	image	Semi-parametric	64×64
CompactSG4LC&PR [72]	scene graph	scene layout	scene layout generation	256×256
Narrative collage [21]	Images	image	narrative collage	——
Scene_generation [6]	scene graph + location information	image	High visual quality	256×256
SIMSG [18]	image + user interacts with the scene graph	image	manipulation	128×128
LostGANs [67]	bounding boxes + object categories	image	layout to image	256×256
CanonicalSg2Im [26]	scene graph	image	scene graph to layout to image	256×256
sg2im-COLR [33]	Scene graph	image	Scene graph to abstract scene to image	128×128

From the above analysis, we observe that realistic-looking, accuracy, operability, and high-resolution are still challenges for text to image generation.

3.2 Scene Graph to Image Generation

Scene Graphs represent scenes as directed graphs, where nodes are objects and edges give relationships between objects.

Table 2 shows that the current scene graph to image generation models can be divided into the following categories: 1. Pure scene graph to image generation, that is, images are directly generated from scene graphs. The representative articles are [6, 34, 42, 71, 93]. 2. To achieve the purpose of manipulable image generation through scene graph manipulation, that is, the manipulation image generation task, the representative articles are [18, 52]. There is also a layout to image generation application similar to the scene graph to image synthesis task. We also list it in the table. The representative article is [67]. Layout sometimes appears as an intermediate result in the scene graph to image generation model, as in article [26]. There are similar articles [33], but the intermediate bridge becomes an abstract scene. Scene graph sometimes acts as an intermediate bridge for image generation, such as in narrative collage task [21]. From the table, we also observe the limitation that the size of the image generated from the current scene graph to image is basically 256×256 or less. Moreover, the training time of scene graph to image generation is always quite long. These are problems that need to be solved in the task of scene graph to image.

3.3 Semantic layout to Image Generation

Image semantic segmentation (semantic layout) simplifies the representation of an image into color segments that are easier to analyze.

The image-to-image transfer models [32, 78, 96] naturally can generate images from semantic labels because semantic labels are a kind of image. Table 3 shows that the current semantic map to image generation models focus on the following challenges: 1, to generate higher-resolution images from semantic labels, such as in [13, 78]; 2, to generate more realistic images from semantic labels, such as in [13]; 3, to make the image generation results controllable, such as in [7, 30, 36, 54, 55, 77, 99], that is, manipulation by change semantic label inputs; 4, to increase the controllable factors in image generation, except semantic labels, you can change image style as well, such as in [54, 77]. In terms of image generation resolution, the semantic layouts to image generation models can generate larger images, and some can even generate large images, such as 1024×2048 resolution.

3.4 Text-based Colorization

Image colorization is an essential branch of image processing and computer vision to colorize images. In text-based colorization deep learning networks, the images are colorized based on the input, which usually are text descriptions.

In [24], the authors propose FiLMedGAN, which leverages feature-wise linear modulation (FiLM) to relate and transform visual features with natural language representations without using extra spatial information. Manjunatha et al. [48] present two different architectures for language-conditioned colorization, both of which produce more accurate and plausible colorizations than a language-agnostic version. Zhang et al. [90] propose a fully automatic approach that makes vibrant and realistic colorizations and embraces the problem's underlying uncertainty by posing it as a classification task and using class-rebalancing at training time to increase the diversity of colors in the result. Bahng et al. [8] propose an approach to generate multiple color palettes that reflect the semantics of input text and then colorize a given grayscale image according to the generated color palette. Kim et al. [37] presented a GAN-based line-art colorization called Tag2Pix, which produces high-quality colored images for given line art and color tags and trained the network with an introduced Tag2Pix dataset. Table 4 shows a summary of these text-based colorization models.

3.5 Multimodal medical Image Generation

In medical imaging, such as magnetic resonance imaging (MRI), you can use different imaging settings (such as T1, T2, DWI, Flair) to obtain several images, multimodal representations of the same image. These multimodal representations (images) have similar anatomical structures but have different contrasts, providing much information for diagnosis. Zhou et al. [95] proposed an end-to-end hybrid-fusion network for multimodal MR image synthesis. Specifically, the method explores the modality-specific properties within each modality and simultaneously exploits the correlations across multiple modalities. Togo et al. [70] present a multimodal

Table 3: A summary of different semantic layout to image generation models

Methods	Input	Output	Characteristics	Resolution
pix2pix [32]	semantic label	image	first pix2pix software	512×512
pix2pixHD [78]	semantic label	image	high-resolution	2048×1024
SPADE [54]	semantic label	image	semantic and style controllable	512×256
AL-CGAN [36]	semantic layout + scene attributes	image	conditional	128×128
Hierchical_image_manipulation [30]	semantic layout + image	image	manipulation	——
Style-semantics [55]	semantic layout + text	image	semantics + style control	256×256
SB-GAN [7]	random noise then semantic labels	image	conditional	128×256
Sequential [74]	semantic layout	image	step by step	128×128
Cycle-GAN [96]	semantic label	image	cycle + a source domain to a target domain	256×256
Photographicimagesynthesis [13]	semantic label	image	photographic result	1024×2048
pix2pixSC [77]	semantic label + style image	image	semantic + style conditional	256×256
Bounding-box-represented [44]	semantic label + bounding box	image	translator enhancer structure	512×512
GroupDNet [99]	semantic label	image	semantic manipulation	512×256

Table 4: A summary of different text-based colorization models

Methods	Input	Output	Characteristics	Resolution
FiLMedGAN [24]	image + text	image	natural language condition	128×64
colorfromlanguage [48]	image + caption	image	caption condition	224×224
colorization [90]	grayscale image	image	like real color phot	256×256
Text2Colors [8]	grayscale image + text	image	multiple palettes	——
Tag2Pix [37]	grayscale image + tags	image	high quality	512×512

image-to-image translation model for the generation of gastritis images using X-ray and blood inspection results. Liu et al. [45] proposed an end-to-end multi-input and multi-output deep adversarial learning framework for MR image synthesis. Chartsias et al. [12] proposed a multi-input, multi-output end-to-end deep convolutional network for the synthesis of MR images, capable of fusing information contained in different modalities. Xiao et al. [84] proposed Transfer-GAN, an end-to-end multimodal image super-resolution network with the transfer learning strategy, which is the first-time for transfer learning and GAN being integrated for multimodal CT image super-resolution. It also provides a potential solution for maintaining high image quality in support of radiation dose optimization in multimodal CT scanning, providing a safer multimodal CT scan strategy for comprehensive brain imaging. Although there are not so many medical image synthesis applications based on multimodal deep learning, it is a convenient and promising application field. Due to medical images' multimodal characteristics, multimodal deep learning image synthesis tasks will shine in medical influence processing in the future.

4 DATASETS

In this section, we review the datasets used in the methods surveyed in this paper. We will use five tables to count the datasets of the five application areas separately: text to image, scene graph to image, semantic layout to image, text-based colorization, and multimodal medical image synthesis.

4.1 Datasets for text to image generation

As shown in Table 5, the text to image task frequently uses twelve datasets: CUB [81], Oxford102 [53], MHP [3], COCO [43], VisDial [16], CelebA [46], AWA2 [83], MULTI-MNIST [20], CLEVR [35], CoDraw [38], DeepFashion [98], and FashionGen [62]. From the table we can find that the top three most commonly used ones are: COCO [43], CUB [81], and Oxford102 [53].

CUB [81] contains 11,788 bird images belonging to 200 categories, and each image is annotated with 10 descriptions. Oxford102 [53] contains 8,189 flow images in 102 categories. Each image is annotated with 10 descriptions too. MHP [3] has 25K images with 410 different common activities. COCO [43] contains 82, 783 training images and 40, 504 validation images. Each image has 5 text annotations. VisDial [16] consists of 10 question-answer conversation turns per dialogue and has one dialogue for each of the MS COCO images. CelebA [46] contains more than 200, 000 celebrity face images, each sample has 40 attribute annotations and the images cover large pose variations and background clutter. AWA2 [83] contains a collection of 37, 322 images for the 50 classes of AWA1 [40]. MULTI-MNIST [20] is a set of multi-MNIST digits. Each image contains zero, one or two non-overlapping random MNIST digits with equal probability. CLEVR [35] is a programmatically generated dataset that is popular in the Visual Question Answering (VQA) community. CLEVR consists of images of collections of objects with different shapes, colors, materials and sizes. Each image is assigned complex questions about object counts, attributes or existence. CoDraw [38] is a recently released clip art-like dataset. It consists of scenes, which are sequences of images of children playing in a park. The children have different poses and expressions and the scenes include other objects such as trees, tables, and animals. There are 58 object types in total. Corresponding to every scene, there is a conversation between a Teller and a Drawer (both Amazon Mechanical Turk workers) in natural language. The Drawer updates the canvas based on the Teller's instructions. The Drawer can ask questions as well for clarification. The dataset consists of 9, 993 scenes of varying length. DeepFashion [98] is a large-scale clothes dataset with comprehensive annotations. It contains over

Table 5: Datasets for text to image (CUB [81], Oxford102 [53], MHP [3], COCO [43], VisDial [16], CelebA [46], AWA2 [83], MULTI-MNIST [20] , CLEVR [35], CoDraw [38], DeepFashion [98], and FashionGen [62])

Methods	CUB	Oxford102	MHP	COCO	VisDial	CelebA	AWA2	MULTI-MNIST	CLEVR	CoDraw	DeepFashion	FashionGen
GAN-INT-CLS [60]	✓	✓										
GAWWN [59]	✓		✓									
ChatPainter [65]				✓	✓							
HDGAN [92]	✓	✓		✓								
TAC-GAN [17]		✓										
AttnGAN [85]	✓			✓								
StackGAN [88]	✓	✓		✓								
StackGAN++ [89]	✓	✓		✓								
alignDRAW [49]				✓								
ControlGAN [41]	✓			✓								
Net2Net [61]				✓		✓	✓					
OP-GAN [29]				✓								
Multi-obj-GAN [28]				✓				✓	✓			
DF-GAN [69]	✓			✓								
DM-GAN [97]	✓			✓								
GeNeVA [19]									✓	✓		
LeicaGAN [56]	✓	✓										
MirrorGAN [57]	✓			✓								
TediGAN [82]	✓			✓		✓						
VICTR [25]				✓								
BilinearGAN [50]	✓	✓									✓	
e-AttnGAN [2]											✓	✓
ISLH-Txt2Im [31]				✓								
FiLMedGAN [24]											✓	

Table 6: Datasets for scene graph to image generation: (COCO-Stuff [11], Visual Genome [39], and CLEVR [35])

Methods	COCO-Stuff [11]	Visual Genome [39]	CLEVR [35]
sg2im [34]	✓	✓	
Layout2Im [93]	✓	✓	
Interactive-sg2im [52]	✓		
USGC2im [71]	✓	✓	
PasteGAN [42]	✓	✓	
CompactSG4LC&PR [72]	✓		
Scene_generation [6]	✓		
SIMSG [18]		✓	✓
LostGANs [67]	✓	✓	
CanonicalSg2Im [26]	✓	✓	✓
sg2im-COLR [33]	✓		

800,000 images, which are richly annotated with massive attributes, clothing landmarks, and correspondence of images taken under different scenarios including store, street snapshot, and consumer. FashionGen [62] is a new dataset of 293, 008 high definition (1360 × 1360 pixels) fashion images paired with item descriptions provided by professional stylists. Each item is photographed from a variety of angles.

4.2 Datasets for scene graph to image generation

As shown in Table 6, the scene graph to image synthesis task frequently uses three datasets: COCO-Stuff [11], Visual Genome [39], and CLEVR [35].

COCO-Stuff [11] is a dataset that augments a subset of the COCO dataset [43] with additional stuff categories. The dataset annotates 40K training and 5K validation images with bounding boxes and segmentation masks for 80 thing categories (people, cars, etc.) and 91 stuff categories (sky, grass, etc.). Visual Genome [39] contains over 100K images where each image has an average of 21 objects, 18 attributes, and 18 pairwise relationships between objects.

4.3 Datasets for semantic layout to image generation

As shown in Table 7, the semantic layout to image synthesis task frequently uses nine datasets: COCO [43], DeepFashion [98], COCO-Stuff [11], Visual Genome [39], Cityscapes [14], CMP facades [75], NYU [66], ADE20K [94], and BDD100k [86].

Cityscapes [14] consists of a various set of video sequences recorded in street from several cities, with high quality pixel-level annotations of 5, 000 frames plus a larger set of 20, 000 weakly annotated frames. CMP facades [75] includes 606 rectified images of facades from various sources, which have been manually annotated. The facades are from different cities around the world and diverse architectural styles. NYU [66] is a dataset of 1, 449 RGBD images, capturing 464 diverse indoor scenes, with detailed annotations. ADE20K [94] has 20, 210 images in the training set, 2, 000 images in the validation set, and 3, 000 images in the testing set. All the images are exhaustively annotated with objects. Many objects are also annotated with their parts. For each object, there is additional information about whether it is occluded or cropped, and other attributes. The images in the validation set are exhaustively annotated with parts, while the part annotations are not exhaustive over the images in the training set. BDD100k [86] is a video dataset with 100K videos and 10 tasks to evaluate the exciting progress of

Table 7: Datasets for semantic layout to image generation (COCO [43], DeepFashion [98], COCO-Stuff [11], Visual Genome [39], Cityscapes [14], CMP facades [75], NYU [66], ADE20K [94], and BDD100k [86])

Methods	COCO	DeepFashion	COCO-Stuff	Visual Genome	Cityscapes	CMP facades	NYU	ADE20K	BDD100k
pix2pix [32]					✓	✓			
pix2pixHD [78]					✓		✓	✓	
SPADE [54]			✓		✓				
AL-CGAN [36]								✓	
Hierchical_image_manipulation [30]					✓			✓	
Style-semantics [55]			✓	✓					
SB-GAN [7]					✓			✓	
Sequential [74]	✓								
Cycle-GAN [96]					✓	✓			
Photographicimagesynthesis [13]					✓		✓		
pix2pixSC [77]									✓
Bounding-box-represented [44]						✓			
GroupDNet [99]		✓			✓			✓	

Table 8: Datasets for text-based colorization (COCO [43], DeepFashion [98], ImageNet [63], PAT [8], and Tag2Pix [37])

Methods	COCO	DeepFashion	ImageNet [63]	PAT [8]	Tag2Pix [37]
FiLMedGAN [24]		✓			
colorfromlanguage [48]	✓				
colorization [90]			✓		
Text2Colors [8]				✓	
Tag2Pix [37]					✓

image recognition algorithms on autonomous driving. The dataset possesses geographic, environmental, and weather diversity, which is useful for training models that are less likely to be surprised by new conditions.

4.4 Datasets for text-based colorization

As shown in Table 8, the text-based colorization task frequently uses five datasets: COCO [43], DeepFashion [98], ImageNet [63], PAT [8], and Tag2Pix [37].

ImageNet [63] is mainly collected to be used for object recognition and contains more than 14 million images in 20, 000 categories, which hand-annotated. PAT [8] contains 10, 183 text and five-color palette pairs, where the set of five colors in a palette is associated with its corresponding text description. Words vary with respect to their relationships with colors; some words are direct color words (e.g., pink, blue, etc.) while others evoke a particular set of colors (e.g., autumn or vibrant). Tag2Pix [37] is a selection of Danbooru2017 [4]. Each image used from the Danbooru2017 set has a size of 512×512 letterboxed.

4.5 Datasets for multimodal medical image synthesis

As shown in Table 9, the multimodal medical image synthesis task frequently uses three datasets: BraTs2018 [51], ISLES [47], and IXI [10].

BraTs2018 [51] consists of 285 patients with multiple MR scans acquired from 19 different institutions and includes glioblastoma (GBM) and lower grade glioma (LGG) cohorts. The patient scans contain four modalities of co-registered MR volumes: T1, T1c, T2, and Flair, where each modality volume is of size $240 \times 240 \times 155$. ISLES [47] is a well described, carefully selected annotated multi812 spectral MRI datasets under a research license, developed a 813 suitable ranking system and invited research groups from all 814 over the world to participate. IXI [10] collects nearly 600 MR images from normal, healthy subjects. The MR image acquisition protocol for each subject includes: T1, T2 and PD-weighted images, MRA images, and Diffusion-weighted images (15 directions).

Table 9: Datasets for multimodal medical image synthesis (BraTs2018 [51], ISLES [47], and IXI [10])

Methods	BraTs2018 [51]	ISLES [47]	IXI [10]
Hi-Net [95]	✓		
Multimodal-MR-Synthesis [45]	✓	✓	
Multimodal_brain_synthesis [12]	✓	✓	✓

5 EVALUATION METRICS

Evaluation metrics can evaluate the quality of generated images, an inseparable part of deep learning.

Table 10 shows the commonly used evaluation metrics (IS [64], MS-SSIM [80], R-precision [15], FID [27], SSIM [79], LPIPS [91], NMSE, and PSNR.) for image generation and the evaluation metrics used by methods reviewed in this paper. As shown from the table, Inception score [64] and FID [27] are the two most commonly used evaluation methods. NMSE is an acronym for normalized mean squared error, and PSNR is an acronym for peak signal-to-noise ratio. IS [64] is an acronym for Inception score, which calculates the conditional distribution entropy by applying the Inception Model [68] and marginal distribution of a large set of generated images. Low entropy of conditional distribution means that the generative model can produce high-quality samples. High entropy of the marginal distribution implies that the generated samples are also

Table 10: Evaluation metrics for multimodal deep learning image synthesis

Methods	IS [64]	MS-SSIM [80]	R-precision [15]	FID [27]	SSIM [79]	LPIPS [91]	NMSE	PSNR
GAN-INT-CLS [60]								
GAWWN [59]								
ChatPainter [65]	✓							
HDGAN [92]	✓							
TAC-GAN [17]	✓	✓						
AttnGAN [85]	✓		✓					
StackGAN [88]	✓							
StackGAN++ [89]	✓			✓				
alignDRAW [49]					✓			
ControlGAN [41]	✓		✓					
Net2Net [61]				✓				
OP-GAN [29]	✓		✓	✓				
Multi-obj-GAN [28]	✓			✓				
DF-GAN [69]	✓			✓				
DM-GAN [97]	✓		✓	✓				
GeNeVA [19]				✓				
LeicaGAN [56]	✓		✓					
MirrorGAN [57]	✓		✓					
TediGAN [82]				✓		✓		
VICTR [25]	✓		✓	✓				
BilinearGAN [50]	✓							
e-AttnGAN [2]	✓		✓					
sg2im [34]	✓							
Layout2Im [93]	✓			✓		✓		
Interactive-sg2im [52]	✓					✓		
PasteGAN [42]	✓			✓		✓		
Scene_generation [6]	✓			✓		✓		
SIMSG [18]	✓			✓		✓		
LostGANs [67]	✓			✓		✓		
CanonicalSg2Im [26]	✓							
sg2im-COLR [33]	✓			✓		✓		
SPADE [54]				✓				
ISLH-Txt2Im [31]	✓							
Style-semantics [55]				✓				
SB-GAN [7]				✓				
Sequential [74]				✓				
pix2pixSC [77]				✓				
GroupDNet [99]				✓		✓		
FiLMedGAN [24]	✓			✓				
Text2Colors [8]								
Tag2Pix [37]				✓				
Hi-Net [95]					✓		✓	✓
Multimodal-MR-Synthesis [45]					✓		✓	✓
Multimodal_brain_synthesis [12]					✓			✓

diverse. The IS score is then computed as the KL-divergence between the two entropies. However, the IS also has disadvantages. If the generative model falls into mode collapse, the IS might still be good while the real case is pretty bad. MS-SSIM [80], multi-scale SSIM, a more advanced form of SSIM, is conducted over multiple scales through a process of multiple stages of sub-sampling. It has been shown to perform equally well or better than SSIM on different image and video databases. R-precision [15] is a common evaluation metric for ranking retrieval results. For a given query topic Q, R-precision is the precision at R, where R is the number of relevant documents for Q. If there are r relevant documents among the top-R retrieved documents, then R-precision is r/R. FID [27], Fréchet Inception Distance, is supposed to improve on the IS [64] by comparing the statistics of generated samples to real samples. Lower FID is better, corresponding to more similar real and generated samples measured by the distance between their

Table 11: A summary of performance of different methods with respect to the four benchmark datasets and two performance metrics: Inception Score (IS) and Fréchet Inception Distance (FID). The methods include ChatPainter, HDGAN, TAC-GAN, AttnGAN, StackGAN, StackGAN++, ControlGAN, Net2Net, OP-GAN, DF-GAN, DM-GAN, LeicaGAN, MirrorGAN, BilinearGAN, sg2im, Layout2Im, PasteGAN, Scene_generation, LostGANs, CanonicalSg2Im, sg2im-COLR, SPADE, ISLH-Txt2Im, and Sequential, which are all using the four benchmark datasets. The four benchmark datasets include MS COCO, COCO STUFF, CUB, and Oxford dataset. A dash indicates that no data was found, and the method with an asterisk at the beginning indicates that the COCO Stuff dataset is used

Methods	Datasets & Metrics					
	MS COCO (*Stuff)		CUB [81]		Oxford [53]	
	IS [64]	FID [27]	IS [64]	FID [27]	IS [64]	FID [27]
ChatPainter [65]	9.74 ± 0.02	-	-	-	-	-
HDGAN [92]	11.86±.18	-	4.15±.05	-	-	-
TAC-GAN [17]	-	-	-	-	3.45 ± 0.05	-
AttnGAN [85]	25.89 ± 0.47	-	4.36 ± .03	-	-	-
StackGAN [88]	8.45 ± 0.03	74.05	3.70 ± .04	51.89	3.20 ± .01	55.28
StackGAN++ [89]	8.30 ± .10	81.59	4.04 ± .05	15.30	3.26 ± .01	48.68
ControlGAN [41]	24.06 ± .60	-	4.58 ± .09	-	-	-
*Net2Net [61]	34.7 ± 0.3	30.63	-	-	-	-
OP-GAN [29]	27.88 ± 0.12	24.70 ± 0.09	-	-	-	-
DF-GAN [69]	-	28.92	4.86 ± 0.04	19.24	-	-
DM-GAN [97]	30.49±0.57	32.64	4.75±0.07	16.09	-	-
LeicaGAN [56]	-	-	5.69±0.06	-	3.80±0.01	-
MirrorGAN [57]	26.47 ± 0.41	-	4.56 ± 0.05	-	-	-
BilinearGAN [50]	-	-	2.76±0.08	-	6.26±0.44	-
sg2im [34]	7.3 ± 0.1	-	-	-	-	-
Layout2Im [93]	9.1 ± 0.1	38.14	-	-	-	-
*PasteGAN [42]	10.2 ± 0.2	38.29	-	-	-	-
*Scene_generation [6]	14.5 ± 0.7	81.0	-	-	-	-
LostGANs [67]	18.01 ± 0.50	42.55	-	-	-	-
CanonicalSg2Im [26]	5.6 ± 0.1	-	-	-	-	-
*sg2im-COLR [33]	-	90.3	-	-	-	-
SPADE [54]	-	22.6	-	-	-	-
ISLH-Txt2Im [31]	11.46 ± 0.09	-	-	-	-	-
Sequential [74]	-	23.2	-	-	-	-

activation distributions. As IS, FID also cannot reasonably handle the overfitting problem. SSIM [79], structural similarity index measure, is a method for measuring the similarity between two images, calculated on various windows of an image and based on three comparison measurements of luminance, contrast, and structure. LPIPS [91], learned perceptual image patch similarity, is a diversity metric for generated images given by a weighted L2 distance between deep features of images. It has been demonstrated to correlate well with human perceptual similarity. How to select a good evaluation metric for image generation models is still a challenging problem. An appropriate evaluation metric ought to differentiate true samples from fake ones, verify mode drop, mode collapse, and detect overfitting. There require better methods to evaluate the quality of the image generation results.

6 METHODS RESULTS COMPARISON

A summary of performance of different methods with respect to the four benchmark datasets (MS COCO [43], COCO STUFF [11], CUB [81], and Oxford [53]) and two performance metrics (IS [64] and FID [27]) is reported in Table 11. We select the two most widely used metrics, the four commonly used datasets, and the methods that use these metrics and methods for comparison. From the table, we can see which method performs best on a certain dataset. For example, DM-GAN [97] has the highest IS value on MS COCO dataset, and a higher IS score is better, so DM-GAN performs the best on MS COCO among the methods listed in the table. Meanwhile, a lower FID score is better, which means SPADE [54] performs the best on MS COCO among the methods listed in the table.

7 CONCLUSION AND DISCUSSION

In this paper, we first formalize the multimodal deep learning image synthesis problem. Then we classify the current image synthesis applications of multimodal deep learning image generation task. We review the architecture of typical models designed for multimodal deep learning image synthesis. We overview typical datasets for multimodal deep learning image generation. We introduce mainstream evaluation metrics of multimodal deep learning image synthesis, and finally, we summarize the performance of different methods. We observe that image synthesis's performance has improved in recent years due to the input of multimodal information. The addition of sentences, semantics, labels, and other information makes image generation more practical and vivid. Simultaneously, multimodal deep learning has begun to play an important role in medical image generation. We can expect that in the foreseeable future, deep learning will affect our lives more deeply. However, more new applications call for more new datasets, new models, new fusion strategies, new loss functions, new evaluation metrics, and so on. We believe this article will attract more power to solve these problems.

REFERENCES

[1] Jorge Agnese, Jonathan Herrera, Haicheng Tao, and Xingquan Zhu. 2020. A survey and taxonomy of adversarial neural networks for text-to-image synthesis. *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.* 10, 4 (2020). https://doi.org/10.1002/widm.1345

[2] Kenan Emir Ak, Joo Hwee Lim, Jo Yew Tham, and Ashraf A. Kassim. 2019. Semantically Consistent Hierarchical Text to Fashion Image Synthesis with an Enhanced-Attentional Generative Adversarial Network. In *2019 IEEE/CVF International Conference on Computer Vision Workshops, ICCV Workshops 2019, Seoul, Korea (South), October 27-28, 2019*. IEEE, 3121–3124. https://doi.org/10.1109/ICCVW.2019.00379

[3] Mykhaylo Andriluka, Leonid Pishchulin, Peter V. Gehler, and Bernt Schiele. 2014. 2D Human Pose Estimation: New Benchmark and State of the Art Analysis. In *2014 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2014, Columbus, OH, USA, June 23-28, 2014*. IEEE Computer Society, 3686–3693. https://doi.org/10.1109/CVPR.2014.471

[4] Anonymous, Danbooru community, Gwern Branwen, and Aaron Gokaslan. 2017. Danbooru2017: A Large-Scale Crowdsourced and Tagged Anime Illustration Dataset. https://www.gwern.net/Danbooru2017. https://www.gwern.net/Danbooru2017 Accessed: DATE.

[5] Saeed Anwar, Salman Khan, and Nick Barnes. 2020. A Deep Journey into Super-resolution: A Survey. *ACM Comput. Surv.* 53, 3 (2020), 60:1–60:34. https://doi.org/10.1145/3390462

[6] Oron Ashual and Lior Wolf. 2019. Specifying Object Attributes and Relations in Interactive Scene Generation. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*. IEEE, 4560–4568. https://doi.org/10.1109/ICCV.2019.00466

[7] Samaneh Azadi, Michael Tschannen, Eric Tzeng, Sylvain Gelly, Trevor Darrell, and Mario Lucic. 2019. Semantic Bottleneck Scene Generation. *CoRR* abs/1911.11357 (2019). arXiv:1911.11357 http://arxiv.org/abs/1911.11357

[8] Hyojin Bahng, Seungjoo Yoo, Wonwoong Cho, David Keetae Park, Ziming Wu, Xiaojuan Ma, and Jaegul Choo. 2018. Coloring with Words: Guiding Image Colorization Through Text-Based Palette Generation. In *Computer Vision - ECCV 2018 - 15th European Conference, Munich, Germany, September 8-14, 2018, Proceedings, Part XII (Lecture Notes in Computer Science)*, Vittorio Ferrari, Martial Hebert, Cristian Sminchisescu, and Yair Weiss (Eds.), Vol. 11216. Springer, 443–459. https://doi.org/10.1007/978-3-030-01258-8_27

[9] Tadas Baltrusaitis, Chaitanya Ahuja, and Louis-Philippe Morency. 2019. Multimodal Machine Learning: A Survey and Taxonomy. *IEEE Trans. Pattern Anal. Mach. Intell.* 41, 2 (2019), 423–443. https://doi.org/10.1109/TPAMI.2018.2798607

[10] brain development. [n.d.]. IXI Dataset. ([n. d.]). http://brain-development.org/ixi-dataset/

[11] Holger Caesar, Jasper R. R. Uijlings, and Vittorio Ferrari. 2018. COCO-Stuff: Thing and Stuff Classes in Context. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 1209–1218. https://doi.org/10.1109/CVPR.2018.00132

[12] Agisilaos Chartsias, Thomas Joyce, Mario Valerio Giuffrida, and Sotirios A. Tsaftaris. 2018. Multimodal MR Synthesis via Modality-Invariant Latent Representation. *IEEE Trans. Medical Imaging* 37, 3 (2018), 803–814. https://doi.org/10.1109/TMI.2017.2764326

[13] Qifeng Chen and Vladlen Koltun. 2017. Photographic Image Synthesis with Cascaded Refinement Networks. In *IEEE International Conference on Computer Vision, ICCV 2017, Venice, Italy, October 22-29, 2017*. IEEE Computer Society, 1520–1529. https://doi.org/10.1109/ICCV.2017.168

[14] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. 2016. The Cityscapes Dataset for Semantic Urban Scene Understanding. In *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016, Las Vegas, NV, USA, June 27-30, 2016*. IEEE Computer Society, 3213–3223. https://doi.org/10.1109/CVPR.2016.350

[15] Nick Craswell. 2009. *R-Precision*. Springer US, Boston, MA, 2453–2453. https://doi.org/10.1007/978-0-387-39940-9_486

[16] Abhishek Das, Satwik Kottur, Khushi Gupta, Avi Singh, Deshraj Yadav, José M. F. Moura, Devi Parikh, and Dhruv Batra. 2017. Visual Dialog. In *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*. IEEE Computer Society, 1080–1089. https://doi.org/10.1109/CVPR.2017.121

[17] Ayushman Dash, John Cristian Borges Gamboa, Sheraz Ahmed, Marcus Liwicki, and Muhammad Zeshan Afzal. 2017. TAC-GAN - Text Conditioned Auxiliary Classifier Generative Adversarial Network. *CoRR* abs/1703.06412 (2017). arXiv:1703.06412 http://arxiv.org/abs/1703.06412

[18] Helisa Dhamo, Azade Farshad, Iro Laina, Nassir Navab, Gregory D. Hager, Federico Tombari, and Christian Rupprecht. 2020. Semantic Image Manipulation Using Scene Graphs. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*. IEEE, 5212–5221. https://doi.org/10.1109/CVPR42600.2020.00526

[19] Alaaeldin El-Nouby, Shikhar Sharma, Hannes Schulz, R. Devon Hjelm, Layla El Asri, Samira Ebrahimi Kahou, Yoshua Bengio, and Graham W. Taylor. 2019. Tell, Draw, and Repeat: Generating and Modifying Images Based on Continual Linguistic Instruction. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*. IEEE, 10303–10311. https://doi.org/10.1109/ICCV.2019.01040

[20] S. M. Ali Eslami, Nicolas Heess, Theophane Weber, Yuval Tassa, David Szepesvari, Koray Kavukcuoglu, and Geoffrey E. Hinton. 2016. Attend, Infer, Repeat: Fast Scene Understanding with Generative Models. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett (Eds.). 3225–3233. https://proceedings.neurips.cc/paper/2016/hash/52947e0ade57a09e4a1386d08f17b656-Abstract.html

[21] Fei Fang, Miao Yi, Hui Feng, Shenghong Hu, and Chunxia Xiao. 2018. Narrative Collage of Image Collections by Scene Graph Recombination. *IEEE Trans. Vis. Comput. Graph.* 24, 9 (2018), 2559–2572. https://doi.org/10.1109/TVCG.2017.2759265

[22] Jing Gao, Peng Li, Zhikui Chen, and Jianing Zhang. 2020. A Survey on Deep Learning for Multimodal Data Fusion. *Neural Comput.* 32, 5 (2020), 829–864. https://doi.org/10.1162/neco_a_01273

[23] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron C. Courville, and Yoshua Bengio. 2014. Generative Adversarial Nets. In *Advances in Neural Information Processing Systems 27: Annual Conference on Neural Information Processing Systems 2014, December 8-13 2014, Montreal, Quebec, Canada*, Zoubin Ghahramani, Max Welling, Corinna Cortes, Neil D. Lawrence, and Kilian Q. Weinberger (Eds.). 2672–2680. https://proceedings.neurips.cc/paper/2014/hash/5ca3e9b122f61f8f06494c97b1afccf3-Abstract.html

[24] Mehmet Günel, Erkut Erdem, and Aykut Erdem. 2018. Language Guided Fashion Image Manipulation with Feature-wise Transformations. *CoRR* abs/1808.04000 (2018). arXiv:1808.04000 http://arxiv.org/abs/1808.04000

[25] Soyeon Caren Han, Siqu Long, Siwen Luo, Kunze Wang, and Josiah Poon. 2020. VICTR: Visual Information Captured Text Representation for Text-to-Vision Multimodal Tasks. In *Proceedings of the 28th International Conference on Computational Linguistics, COLING 2020, Barcelona, Spain (Online), December 8-13, 2020*, Donia Scott, Núria Bel, and Chengqing Zong (Eds.). International Committee on Computational Linguistics, 3107–3117. https://doi.org/10.18653/v1/2020.coling-main.277

[26] Roei Herzig, Amir Bar, Huijuan Xu, Gal Chechik, Trevor Darrell, and Amir Globerson. 2020. Learning Canonical Representations for Scene Graph to Image Generation. In *Computer Vision - ECCV 2020 - 16th European Conference, Glasgow, UK, August 23-28, 2020, Proceedings, Part XXVI (Lecture Notes in Computer Science)*, Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm (Eds.), Vol. 12371. Springer, 210–227. https://doi.org/10.1007/978-3-030-58574-7_13

[27] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (Eds.). 6626–6637. https://proceedings.neurips.cc/paper/2017/hash/8a1d694707eb0fefe65871369074926d-Abstract.html

[28] Tobias Hinz, Stefan Heinrich, and Stefan Wermter. 2019. Generating Multiple Objects at Spatially Distinct Locations. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net. https://openreview.net/forum?id=H1edIiA9KQ

[29] Tobias Hinz, Stefan Heinrich, and Stefan Wermter. 2019. Semantic Object Accuracy for Generative Text-to-Image Synthesis. *CoRR* abs/1910.13321 (2019). arXiv:1910.13321 http://arxiv.org/abs/1910.13321

[30] Seunghoon Hong, Xinchen Yan, Thomas E. Huang, and Honglak Lee. 2018. Learning Hierarchical Semantic Image Manipulation through Structured Representations. In *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, Samy Bengio, Hanna M. Wallach, Hugo Larochelle, Kristen Grauman, Nicolò Cesa-Bianchi, and Roman Garnett (Eds.). 2713–2723. https://proceedings.neurips.cc/paper/2018/hash/602d1305678a8d5fdb372271e980da6a-Abstract.html

[31] Seunghoon Hong, Dingdong Yang, Jongwook Choi, and Honglak Lee. 2018. Inferring Semantic Layout for Hierarchical Text-to-Image Synthesis. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 7986–7994. https://doi.org/10.1109/CVPR.2018.00833

[32] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. 2017. Image-to-Image Translation with Conditional Adversarial Networks. In *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*. IEEE Computer Society, 5967–5976. https://doi.org/10.1109/CVPR.2017.632

[33] Maor Ivgi, Yaniv Benny, Avichai Ben-David, Jonathan Berant, and Lior Wolf. 2020. Scene Graph to Image Generation with Contextualized Object Layout Refinement.

CoRR abs/2009.10939 (2020). arXiv:2009.10939 https://arxiv.org/abs/2009.10939

[34] Justin Johnson, Agrim Gupta, and Li Fei-Fei. 2018. Image Generation From Scene Graphs. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 1219–1228. https://doi.org/10.1109/CVPR.2018.00133

[35] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross B. Girshick. 2017. CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning. In *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*. IEEE Computer Society, 1988–1997. https://doi.org/10.1109/CVPR.2017.215

[36] Levent Karacan, Zeynep Akata, Aykut Erdem, and Erkut Erdem. 2016. Learning to Generate Images of Outdoor Scenes from Attributes and Semantic Layouts. *CoRR* abs/1612.00215 (2016). arXiv:1612.00215 http://arxiv.org/abs/1612.00215

[37] Hyunsu Kim, Ho Young Jhoo, Eunhyeok Park, and Sungjoo Yoo. 2019. Tag2Pix: Line Art Colorization Using Text Tag With SECat and Changing Loss. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*. IEEE, 9055–9064. https://doi.org/10.1109/ICCV.2019.00915

[38] Jin-Hwa Kim, Nikita Kitaev, Xinlei Chen, Marcus Rohrbach, Byoung-Tak Zhang, Yuandong Tian, Dhruv Batra, and Devi Parikh. 2019. CoDraw: Collaborative Drawing as a Testbed for Grounded Goal-driven Communication. In *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, Anna Korhonen, David R. Traum, and Lluís Màrquez (Eds.). Association for Computational Linguistics, 6495–6513. https://doi.org/10.18653/v1/p19-1651

[39] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A. Shamma, Michael S. Bernstein, and Li Fei-Fei. 2017. Visual Genome: Connecting Language and Vision Using Crowdsourced Dense Image Annotations. *Int. J. Comput. Vis.* 123, 1 (2017), 32–73. https://doi.org/10.1007/s11263-016-0981-7

[40] Christoph H. Lampert, Hannes Nickisch, and Stefan Harmeling. 2014. Attribute-Based Classification for Zero-Shot Visual Object Categorization. *IEEE Trans. Pattern Anal. Mach. Intell.* 36, 3 (2014), 453–465. https://doi.org/10.1109/TPAMI.2013.140

[41] Bowen Li, Xiaojuan Qi, Thomas Lukasiewicz, and Philip H. S. Torr. 2019. Controllable Text-to-Image Generation. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d'Alché-Buc, Emily B. Fox, and Roman Garnett (Eds.). 2063–2073. https://proceedings.neurips.cc/paper/2019/hash/1d72310edc006dadf2190caad5802983-Abstract.html

[42] Yikang Li, Tao Ma, Yeqi Bai, Nan Duan, Sining Wei, and Xiaogang Wang. 2019. PasteGAN: A Semi-Parametric Method to Generate Image from Scene Graph. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d'Alché-Buc, Emily B. Fox, and Roman Garnett (Eds.). 3950–3960. https://proceedings.neurips.cc/paper/2019/hash/12b1e42dc0746f22cf361267de07073f-Abstract.html

[43] Tsung-Yi Lin, Michael Maire, Serge J. Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. 2014. Microsoft COCO: Common Objects in Context. In *Computer Vision - ECCV 2014 - 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V (Lecture Notes in Computer Science)*, David J. Fleet, Tomás Pajdla, Bernt Schiele, and Tinne Tuytelaars (Eds.), Vol. 8693. Springer, 740–755. https://doi.org/10.1007/978-3-319-10602-1_48

[44] Congying Liu, Zexi Yang, Feng Xu, and Jun-Hai Yong. 2019. Image generation from bounding box-represented semantic labels. *Comput. Graph.* 81 (2019), 32–40. https://doi.org/10.1016/j.cag.2019.03.009

[45] Xiaoming Liu, Aihui Yu, Xiangkai Wei, Zhifang Pan, and Jinshan Tang. 2020. Multimodal MR Image Synthesis Using Gradient Prior and Adversarial Learning. *IEEE J. Sel. Top. Signal Process.* 14, 6 (2020), 1176–1188. https://doi.org/10.1109/JSTSP.2020.3013418

[46] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep Learning Face Attributes in the Wild. In *2015 IEEE International Conference on Computer Vision, ICCV 2015, Santiago, Chile, December 7-13, 2015*. IEEE Computer Society, 3730–3738. https://doi.org/10.1109/ICCV.2015.425

[47] Oskar Maier, Bjoern H. Menze, Janina von der Gablentz, Levin Häni, Mattias P. Heinrich, Matthias Liebrand, Stefan Winzeck, Abdul Basit, Paul Bentley, Liang Chen, Daan Christiaens, Francis Dutil, Karl Egger, Chaolu Feng, Ben Glocker, Michael Götz, Tom Haeck, Hanna-Leena Halme, Mohammad Havaei, Khan M. Iftekharuddin, Pierre-Marc Jodoin, and et al. 2017. ISLES 2015 - A public evaluation benchmark for ischemic stroke lesion segmentation from multispectral MRI. *Medical Image Anal.* 35 (2017), 250–269. https://doi.org/10.1016/j.media.2016.07.009

[48] Varun Manjunatha, Mohit Iyyer, Jordan L. Boyd-Graber, and Larry S. Davis. 2018. Learning to Color from Language. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT, New Orleans, Louisiana, USA, June 1-6, 2018, Volume 2 (Short Papers)*, Marilyn A. Walker, Heng Ji, and Amanda Stent (Eds.). Association for Computational Linguistics, 764–769. https://doi.org/10.18653/v1/n18-2120

[49] Elman Mansimov, Emilio Parisotto, Lei Jimmy Ba, and Ruslan Salakhutdinov. 2016. Generating Images from Captions with Attention. In *4th International Conference on Learning Representations, ICLR 2016, San Juan, Puerto Rico, May 2-4, 2016, Conference Track Proceedings*, Yoshua Bengio and Yann LeCun (Eds.). http://arxiv.org/abs/1511.02793

[50] Xiaofeng Mao, Yuefeng Chen, Yuhong Li, Tao Xiong, Yuan He, and Hui Xue. 2019. Bilinear Representation for Language-based Image Editing Using Conditional Generative Adversarial Networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2019, Brighton, United Kingdom, May 12-17, 2019*. IEEE, 2047–2051. https://doi.org/10.1109/ICASSP.2019.8683008

[51] Bjoern H. Menze, András Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin S. Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, Levente Lanczi, Elizabeth R. Gerstner, Marc-André Weber, Tal Arbel, Brian B. Avants, Nicholas Ayache, Patricia Buendia, D. Louis Collins, Nicolas Cordier, Jason J. Corso, Antonio Criminisi, Tilak Das, Herve Delingette, Çagatay Demiralp, Christopher R. Durst, Michel Dojat, Senan Doyle, Joana Festa, Florence Forbes, Ezequiel Geremia, Ben Glocker, Polina Golland, Xiaotao Guo, Andac Hamamci, Khan M. Iftekharuddin, Raj Jena, Nigel M. John, Ender Konukoglu, Danial Lashkari, José Antonio Mariz, Raphael Meier, Sérgio Pereira, Doina Precup, Stephen J. Price, Tammy Riklin Raviv, Syed M. S. Reza, Michael T. Ryan, Duygu Sarikaya, Lawrence H. Schwartz, Hoo-Chang Shin, Jamie Shotton, Carlos A. Silva, Nuno Sousa, Nagesh K. Subbanna, Gábor Székely, Thomas J. Taylor, Owen M. Thomas, Nicholas J. Tustison, Gözde B. Ünal, Flor Vasseur, Max Wintermark, Dong Hye Ye, Liang Zhao, Binsheng Zhao, Darko Zikic, Marcel Prastawa, Mauricio Reyes, and Koen Van Leemput. 2015. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE Trans. Medical Imaging* 34, 10 (2015), 1993–2024. https://doi.org/10.1109/TMI.2014.2377694

[52] Gaurav Mittal, Shubham Agrawal, Anuva Agarwal, Sushant Mehta, and Tanya Marwah. 2019. Interactive Image Generation Using Scene Graphs. In *Deep Generative Models for Highly Structured Data, ICLR 2019 Workshop, New Orleans, Louisiana, United States, May 6, 2019*. OpenReview.net. https://openreview.net/forum?id=SJx-SULKOV

[53] Maria-Elena Nilsback and Andrew Zisserman. 2008. Automated Flower Classification over a Large Number of Classes. In *Sixth Indian Conference on Computer Vision, Graphics & Image Processing, ICVGIP 2008, Bhubaneswar, India, 16-19 December 2008*. IEEE Computer Society, 722–729. https://doi.org/10.1109/ICVGIP.2008.47

[54] Taesung Park, Ming-Yu Liu, Ting-Chun Wang, and Jun-Yan Zhu. 2019. Semantic Image Synthesis With Spatially-Adaptive Normalization. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 2337–2346. https://doi.org/10.1109/CVPR.2019.00244

[55] Dario Pavllo, Aurélien Lucchi, and Thomas Hofmann. 2020. Controlling Style and Semantics in Weakly-Supervised Image Generation. In *Computer Vision - ECCV 2020 - 16th European Conference, Glasgow, UK, August 23-28, 2020, Proceedings, Part VI (Lecture Notes in Computer Science)*, Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm (Eds.), Vol. 12351. Springer, 482–499. https://doi.org/10.1007/978-3-030-58539-6_29

[56] Tingting Qiao, Jing Zhang, Duanqing Xu, and Dacheng Tao. 2019. Learn, Imagine and Create: Text-to-Image Generation from Prior Knowledge. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d'Alché-Buc, Emily B. Fox, and Roman Garnett (Eds.). 885–895. https://proceedings.neurips.cc/paper/2019/hash/d18f655c3fce66ca401d5f38b48c89af-Abstract.html

[57] Tingting Qiao, Jing Zhang, Duanqing Xu, and Dacheng Tao. 2019. MirrorGAN: Learning Text-To-Image Generation by Redescription. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 1505–1514. https://doi.org/10.1109/CVPR.2019.00160

[58] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, and Scott GrayTadas. 2021. DALL·E: Creating Images from Text. (2021). https://openai.com/blog/dall-e/

[59] Scott E. Reed, Zeynep Akata, Santosh Mohan, Samuel Tenka, Bernt Schiele, and Honglak Lee. 2016. Learning What and Where to Draw. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett (Eds.). 217–225. https://proceedings.neurips.cc/paper/2016/hash/a8f15eda80c50adb0e71943adc8015cf-Abstract.html

[60] Scott E. Reed, Zeynep Akata, Xinchen Yan, Lajanugen Logeswaran, Bernt Schiele, and Honglak Lee. 2016. Generative Adversarial Text to Image Synthesis. In *Proceedings of the 33nd International Conference on Machine Learning, ICML 2016,*

New York City, NY, USA, June 19-24, 2016 (JMLR Workshop and Conference Proceedings), Maria-Florina Balcan and Kilian Q. Weinberger (Eds.), Vol. 48. JMLR.org, 1060–1069. http://proceedings.mlr.press/v48/reed16.html

[61] Robin Rombach, Patrick Esser, and Björn Ommer. 2020. Network-to-Network Translation with Conditional Invertible Neural Networks. In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (Eds.). https://proceedings.neurips.cc/paper/2020/hash/1cfa81af29c6f2d8cacb44921722e753-Abstract.html

[62] Negar Rostamzadeh, Seyedarian Hosseini, Thomas Boquet, Wojciech Stokowiec, Ying Zhang, Christian Jauvin, and Chris Pal. 2018. Fashion-Gen: The Generative Fashion Dataset and Challenge. *CoRR* abs/1806.08317 (2018). arXiv:1806.08317 http://arxiv.org/abs/1806.08317

[63] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael S. Bernstein, Alexander C. Berg, and Fei-Fei Li. 2015. ImageNet Large Scale Visual Recognition Challenge. *Int. J. Comput. Vis.* 115, 3 (2015), 211–252. https://doi.org/10.1007/s11263-015-0816-y

[64] Tim Salimans, Ian J. Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. 2016. Improved Techniques for Training GANs. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett (Eds.). 2226–2234. https://proceedings.neurips.cc/paper/2016/hash/8a3363abe792db2d8761d6403605aeb7-Abstract.html

[65] Shikhar Sharma, Dendi Suhubdy, Vincent Michalski, Samira Ebrahimi Kahou, and Yoshua Bengio. 2018. ChatPainter: Improving Text to Image Generation using Dialogue. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Workshop Track Proceedings*. OpenReview.net. https://openreview.net/forum?id=rJsnEYJwG

[66] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. 2012. Indoor Segmentation and Support Inference from RGBD Images. In *Computer Vision - ECCV 2012 - 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part V (Lecture Notes in Computer Science)*, Andrew W. Fitzgibbon, Svetlana Lazebnik, Pietro Perona, Yoichi Sato, and Cordelia Schmid (Eds.), Vol. 7576. Springer, 746–760. https://doi.org/10.1007/978-3-642-33715-4_54

[67] Wei Sun and Tianfu Wu. 2020. Learning Layout and Style Reconfigurable GANs for Controllable Image Synthesis. *CoRR* abs/2003.11571 (2020). arXiv:2003.11571 https://arxiv.org/abs/2003.11571

[68] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. 2016. Rethinking the Inception Architecture for Computer Vision. In *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016, Las Vegas, NV, USA, June 27-30, 2016*. IEEE Computer Society, 2818–2826. https://doi.org/10.1109/CVPR.2016.308

[69] Ming Tao, Hao Tang, Songsong Wu, Nicu Sebe, Fei Wu, and Xiao-Yuan Jing. 2020. DF-GAN: Deep Fusion Generative Adversarial Networks for Text-to-Image Synthesis. *CoRR* abs/2008.05865 (2020). arXiv:2008.05865 https://arxiv.org/abs/2008.05865

[70] Ren Togo, Takahiro Ogawa, and Miki Haseyama. 2020. Multimodal Image-to-Image Translation for Generation of Gastritis Images. In *IEEE International Conference on Image Processing, ICIP 2020, Abu Dhabi, United Arab Emirates, October 25-28, 2020*. IEEE, 2466–2470. https://doi.org/10.1109/ICIP40778.2020.9190863

[71] Subarna Tripathi, Anahita Bhiwandiwalla, Alexei Bastidas, and Hanlin Tang. 2019. Using Scene Graph Context to Improve Image Generation. *CoRR* abs/1901.03762 (2019). arXiv:1901.03762 http://arxiv.org/abs/1901.03762

[72] Subarna Tripathi, Sharath Nittur Sridhar, Sairam Sundaresan, and Hanlin Tang. 2019. Compact Scene Graphs for Layout Composition and Patch Retrieval. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops, CVPR Workshops 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 676–683. https://doi.org/10.1109/CVPRW.2019.00094

[73] Apostolia Tsirikoglou, Gabriel Eilertsen, and Jonas Unger. 2020. A Survey of Image Synthesis Methods for Visual Machine Learning. *Comput. Graph. Forum* 39, 6 (2020), 426–451. https://doi.org/10.1111/cgf.14047

[74] Mehmet Ozgur Turkoglu, William Thong, Luuk J. Spreeuwers, and Berkay Kicanaoglu. 2019. A Layer-Based Sequential Framework for Scene Generation with GANs. In *The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019, The Thirty-First Innovative Applications of Artificial Intelligence Conference, IAAI 2019, The Ninth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019, Honolulu, Hawaii, USA, January 27 - February 1, 2019*. AAAI Press, 8901–8908. https://doi.org/10.1609/aaai.v33i01.33018901

[75] Radim Tylecek and Radim Sára. 2013. Spatial Pattern Templates for Recognition of Objects with Regular Structure. In *Pattern Recognition - 35th German Conference, GCPR 2013, Saarbrücken, Germany, September 3-6, 2013. Proceedings (Lecture Notes in Computer Science)*, Joachim Weickert, Matthias Hein, and Bernt Schiele (Eds.), Vol. 8142. Springer, 364–374. https://doi.org/10.1007/978-3-642-40602-7_39

[76] Lei Wang, Wei Chen, Wenjia Yang, Fangming Bi, and Fei Richard Yu. 2020. A State-of-the-Art Review on Image Synthesis With Generative Adversarial Networks. *IEEE Access* 8 (2020), 63514–63537. https://doi.org/10.1109/ACCESS.2020.2982224

[77] Miao Wang, Guo-Ye Yang, Ruilong Li, Runze Liang, Song-Hai Zhang, Peter M. Hall, and Shi-Min Hu. 2019. Example-Guided Style-Consistent Image Synthesis From Semantic Labeling. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 1495–1504. https://doi.org/10.1109/CVPR.2019.00159

[78] Ting-Chun Wang, Ming-Yu Liu, Jun-Yan Zhu, Andrew Tao, Jan Kautz, and Bryan Catanzaro. 2018. High-Resolution Image Synthesis and Semantic Manipulation With Conditional GANs. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 8798–8807. https://doi.org/10.1109/CVPR.2018.00917

[79] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. 2004. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Process.* 13, 4 (2004), 600–612. https://doi.org/10.1109/TIP.2003.819861

[80] Zhou Wang, Eero P Simoncelli, and Alan C Bovik. 2003. Multiscale structural similarity for image quality assessment. In *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, Vol. 2. Ieee, 1398–1402.

[81] Peter Welinder, Steve Branson, Takeshi Mita, Catherine Wah, Florian Schroff, Serge Belongie, and Pietro Perona. 2010. Caltech-UCSD birds 200. (2010).

[82] Weihao Xia, Yujiu Yang, Jing-Hao Xue, and Baoyuan Wu. 2020. TediGAN: Text-Guided Diverse Image Generation and Manipulation. *CoRR* abs/2012.03308 (2020). arXiv:2012.03308 https://arxiv.org/abs/2012.03308

[83] Yongqin Xian, Christoph H. Lampert, Bernt Schiele, and Zeynep Akata. 2019. Zero-Shot Learning - A Comprehensive Evaluation of the Good, the Bad and the Ugly. *IEEE Trans. Pattern Anal. Mach. Intell.* 41, 9 (2019), 2251–2265. https://doi.org/10.1109/TPAMI.2018.2857768

[84] Yao Xiao, Keith R. Peters, W. Christopher Fox, John H. Rees, Dhanashree A. Rajderkar, Manuel M. Arreola, Izabella Barreto, Wesley E. Bolch, and Ruogu Fang. 2020. Transfer-Gan: Multimodal Ct Image Super-Resolution Via Transfer Generative Adversarial Networks. In *17th IEEE International Symposium on Biomedical Imaging, ISBI 2020, Iowa City, IA, USA, April 3-7, 2020*. IEEE, 195–198. https://doi.org/10.1109/ISBI45749.2020.9098322

[85] Tao Xu, Pengchuan Zhang, Qiuyuan Huang, Han Zhang, Zhe Gan, Xiaolei Huang, and Xiaodong He. 2018. AttnGAN: Fine-Grained Text to Image Generation With Attentional Generative Adversarial Networks. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 1316–1324. https://doi.org/10.1109/CVPR.2018.00143

[86] Fisher Yu, Haofeng Chen, Xin Wang, Wenqi Xian, Yingying Chen, Fangchen Liu, Vashisht Madhavan, and Trevor Darrell. 2020. BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*. IEEE, 2633–2642. https://doi.org/10.1109/CVPR42600.2020.00271

[87] Chao Zhang, Zichao Yang, Xiaodong He, and Li Deng. 2020. Multimodal Intelligence: Representation Learning, Information Fusion, and Applications. *IEEE J. Sel. Top. Signal Process.* 14, 3 (2020), 478–493. https://doi.org/10.1109/JSTSP.2020.2987728

[88] Han Zhang, Tao Xu, and Hongsheng Li. 2017. StackGAN: Text to Photo-Realistic Image Synthesis with Stacked Generative Adversarial Networks. In *IEEE International Conference on Computer Vision, ICCV 2017, Venice, Italy, October 22-29, 2017*. IEEE Computer Society, 5908–5916. https://doi.org/10.1109/ICCV.2017.629

[89] Han Zhang, Tao Xu, Hongsheng Li, Shaoting Zhang, Xiaogang Wang, Xiaolei Huang, and Dimitris N. Metaxas. 2019. StackGAN++: Realistic Image Synthesis with Stacked Generative Adversarial Networks. *IEEE Trans. Pattern Anal. Mach. Intell.* 41, 8 (2019), 1947–1962. https://doi.org/10.1109/TPAMI.2018.2856256

[90] Richard Zhang, Phillip Isola, and Alexei A. Efros. 2016. Colorful Image Colorization. In *Computer Vision - ECCV 2016 - 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part III (Lecture Notes in Computer Science)*, Bastian Leibe, Jiri Matas, Nicu Sebe, and Max Welling (Eds.), Vol. 9907. Springer, 649–666. https://doi.org/10.1007/978-3-319-46487-9_40

[91] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 586–595. https://doi.org/10.1109/CVPR.2018.00068

[92] Zizhao Zhang, Yuanpu Xie, and Lin Yang. 2018. Photographic Text-to-Image Synthesis With a Hierarchically-Nested Adversarial Network. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. IEEE Computer Society, 6199–6208. https://doi.org/10.1109/CVPR.2018.00649

[93] Bo Zhao, Lili Meng, Weidong Yin, and Leonid Sigal. 2019. Image Generation From Layout. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 8584–8593. https://doi.org/10.1109/CVPR.2019.00878

[94] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. 2017. Scene Parsing through ADE20K Dataset. In *2017 IEEE Conference*

on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017. IEEE Computer Society, 5122–5130. https://doi.org/10.1109/CVPR.2017.544

[95] Tao Zhou, Huazhu Fu, Geng Chen, Jianbing Shen, and Ling Shao. 2020. Hi-Net: Hybrid-Fusion Network for Multi-Modal MR Image Synthesis. *IEEE Trans. Medical Imaging* 39, 9 (2020), 2772–2781. https://doi.org/10.1109/TMI.2020.2975344

[96] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. 2017. Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks. In *IEEE International Conference on Computer Vision, ICCV 2017, Venice, Italy, October 22-29, 2017*. IEEE Computer Society, 2242–2251. https://doi.org/10.1109/ICCV.2017.244

[97] Minfeng Zhu, Pingbo Pan, Wei Chen, and Yi Yang. 2019. DM-GAN: Dynamic Memory Generative Adversarial Networks for Text-To-Image Synthesis. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 5802–5810. https://doi.org/10.1109/CVPR.2019.00595

[98] Shizhan Zhu, Sanja Fidler, Raquel Urtasun, Dahua Lin, and Chen Change Loy. 2017. Be Your Own Prada: Fashion Synthesis with Structural Coherence. In *IEEE International Conference on Computer Vision, ICCV 2017, Venice, Italy, October 22-29, 2017*. IEEE Computer Society, 1689–1697. https://doi.org/10.1109/ICCV.2017.186

[99] Zhen Zhu, Zhiliang Xu, Ansheng You, and Xiang Bai. 2020. Semantically Multi-Modal Image Synthesis. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*. IEEE, 5466–5475. https://doi.org/10.1109/CVPR42600.2020.00551